Lab 14 — Enforce Standards with Azure Policy

In this lab you use Azure Policy to enforce an organisational standard — restricting which regions resources may be deployed to — and watch a non-compliant deployment get denied at creation time, even for an Owner. This maps to the AZ-900 skill area "Describe Azure management and governance" (purpose of Azure Policy and compliance).

Run on https://portal.azure.com — sign in with your Azure free account (https://azure.microsoft.com/free).

Step 1 — Open the Policy service

1. In the portal search bar, type **Policy** and open the **Policy** service.
2. On the **Overview** blade, note the compliance summary for your subscription (likely 100% or "no assignments" if this is a fresh account).

Azure Policy evaluates every resource against assigned rules and reports compliance continuously — it is governance for *what* gets created, not *who* creates it.

Step 2 — Browse built-in policy definitions

1. In the Policy left menu, under **Authoring**, click **Definitions**.
2. In the **Search** box type **Allowed locations** — open the built-in definition named **Allowed locations**.
3. Read the description and click the **JSON** view: the rule denies any resource whose `location` is not in the `listOfAllowedLocations` parameter.
4. Go back and search **Require a tag on resources** — open it and note it denies creation of resources missing a given tag.

Azure ships hundreds of built-in definitions (many grouped into **initiatives** — bundles of related policies); you assign them rather than writing JSON from scratch.

Step 3 — Assign "Allowed locations" to rg-az900-labs

1. From the **Allowed locations** definition page, click **Assign policy** (or go to **Assignments** > **Assign policy** and pick the definition).
2. On the **Basics** tab:

- **Scope**: click the `...` button, select your subscription, then **Resource Group**: `rg-az900-labs`, click **Select**.
- **Assignment name**: leave as `Allowed locations`.

1. On the **Parameters** tab:

- **Allowed locations**: tick **Southeast Asia** and **East Asia** only.

1. Click **Review + create** > **Create**.

The scope matters: this rule now applies only inside `rg-az900-labs`; assigned at subscription or management-group scope it would govern everything below.

Step 4 — Wait for the assignment to take effect

1. Go to **Policy** > **Assignments** and confirm the **Allowed locations** assignment is listed with scope `rg-az900-labs`.
2. Wait about **5 minutes** before testing — new assignments take a few minutes to propagate to the resource providers that enforce them.

Enforcement (deny at deployment) starts within minutes; the **Compliance** dashboard scan is slower (see Step 6).

Step 5 — Attempt a non-compliant deployment

1. Open **rg-az900-labs** > **+ Create**, search **Storage account**, click **Create**.
2. On the **Basics** tab:

- **Resource group**: `rg-az900-labs`
- **Storage account name**: `staz900policytest` plus random digits (lowercase, unique)
- **Region**: `West Europe` < deliberately not allowed
- **Redundancy**: `Locally-redundant storage (LRS)`

1. Click **Review + create**. Validation fails (or the deployment is denied on **Create**) with an error like:

- `Resource 'staz900policytest...' was disallowed by policy. Policy identifiers: '[{"policyAssignment":{"name":"Allowed locations", ...}}]' (Code: RequestDisallowedByPolicy)`

1. Read the error — it names the exact **policy assignment** that blocked you. Cancel the deployment (do not retry with an allowed region).

You are the subscription Owner and were still denied: Azure Policy applies to **everyone**, regardless of RBAC role.

Step 6 — View the Compliance blade

1. Go to **Policy** > **Compliance**.
2. Click the **Allowed locations** assignment to see its compliance state and the count of compliant resources (your existing Southeast Asia resources comply).

Note: the compliance evaluation cycle can take **~15 minutes or more** to first populate — if it shows "Not started", continue and check back later. Denied deployments never appear here because the resource was never created.

Step 7 — Compare Azure Policy with RBAC

	Azure RBAC (Lab 12)	Azure Policy (this lab)
Question answered	**Who** can perform actions?	**What** configurations are allowed?
Applies to	The assigned principal only	Everyone, including Owners
Example	Reader cannot create resources	No one can create resources in West Europe
Blocked how	Authorization error before the action	`RequestDisallowedByPolicy` on deployment / flagged non-compliant

Real governance uses both together: RBAC limits who acts, Policy limits what even authorised users can do.

Step 8 — Delete the policy assignment

1. Go to **Policy** > **Assignments**.
2. Click the `...` at the right of the **Allowed locations** assignment and select **Delete assignment**, then confirm.

Or with the CLI in Cloud Shell:

```
az policy assignment list --resource-group rg-az900-labs -o table
az policy assignment delete --name "Allowed locations" --resource-group
rg-az900-labs
```

The first command shows the assignment's name at the resource-group scope; the second removes it so later labs can deploy to any region.

Clean up

1. Confirm the **Allowed locations** assignment is deleted (Step 8) — the built-in *definition* stays, only your *assignment* is removed.
2. No resources were created (the test deployment was denied), so nothing else to delete. Keep `rg-az900-labs` for the remaining labs.

What you learned

- Azure Policy enforces organisational standards (allowed regions, required tags, allowed SKUs) and continuously assesses **compliance**.
- A policy is a **definition** (the rule) plus an **assignment** (the rule applied at a scope); initiatives bundle related definitions.
- Policy assignments inherit down the scope hierarchy just like RBAC, but apply to **all users including Owners**.
- Deny-effect policies block non-compliant deployments with `RequestDisallowedByPolicy`, naming the assignment responsible.
- RBAC answers *who can act*; Policy answers *what is allowed* — the AZ-900 exam tests this distinction.

Files in this lab folder

File	What it is
`allowed-locations-policy.json`	Parameters file for the built-in **Allowed locations** policy.
`require-tag-policy.json`	A custom policy rule that denies any resource created without a `course` tag.
`POLICY-USAGE.md`	How to assign, test and remove both policies from Cloud Shell.

The scripts are optional conveniences — the lab steps above teach the same thing command by command. Upload a script to Cloud Shell (**Manage files > Upload**), or clone this repo there, then run it with `bash <script-name>`.